Experiment 10

Aim: To implement a Hyperledger Fabric blockchain network and perform asset management operations such as creating, reading, updating, and transferring assets using chaincode.

Theory: Hyperledger Fabric is an open-source, permissioned blockchain framework designed for enterprise applications. Unlike public blockchains, it allows only authorized participants to join the network. It provides privacy, security, transparency, and tamper-resistant transaction records.

The main components of Hyperledger Fabric are:

1. Peer Nodes: Maintain the ledger and execute chaincode.
2. Ordering Service: Orders transactions and distributes blocks to peers.
3. Channel: Provides a private communication and transaction space for participating organizations.
4. Ledger: Stores the blockchain transaction history and the current state of assets.
5. Chaincode: A smart contract containing business logic for asset management.
6. Certificate Authority (CA): Manages digital identities and certificates for network participants.

Asset Management using Chaincode

Asset management involves maintaining information about digital or physical assets on a blockchain. Chaincode defines the operations that users can perform on these assets.

The following operations are performed in this experiment:

- Create Asset: Adds a new asset to the ledger.
- Read Asset: Retrieves asset details from the ledger.
- Update Asset: Modifies the information associated with an asset.

- Transfer Asset: Changes the ownership of an asset.
- Delete Asset: Removes an asset from the current ledger state, if permitted by the chaincode.

Hyperledger Fabric uses a distributed ledger, ensuring that authorized participants can verify transactions and that changes are recorded in a traceable manner.

Procedure

1. Install and configure Docker Desktop and Git.
2. Download the Hyperledger Fabric samples, binaries, and Docker images.
3. Navigate to the Fabric test network directory.
4. Start the Fabric test network using Docker containers.
5. Create a channel for communication between network participants.
6. Deploy the asset-management chaincode on the network.
7. Invoke the chaincode to create a new asset.
8. Query the ledger to retrieve the asset details.
9. Update the asset information and transfer ownership.
10. Verify the updated asset information and capture screenshots of the outputs.

Step 1: Check prerequisites

```
docker --version%
(base) deepakadvani@deepaks-MacBook ~ % docker --version
Docker version 29.3.1, build c2be9cc
(base) deepakadvani@deepaks-MacBook ~ % docker info
```

```
(base) deepakadvani@deepaks-MacBook ~ % git --version
git version 2.50.1 (Apple Git-155)
(base) deepakadvani@deepaks-MacBook ~ % uname -m
arm64
(base) deepakadvani@deepaks-MacBook ~ %
```

Step 2: Download Hyperledger Fabric

```
Last login: Thu Nov  5 13:29:07 on console
(base) deepakadvani@deepaks-MacBook ~ % uname -m
arm64
(base) deepakadvani@deepaks-MacBook ~ % cd ~/HyperledgerExperiment10
(base) deepakadvani@deepaks-MacBook HyperledgerExperiment10 % ls -lah
total 24
drwxr-xr-x    4 deepakadvani  staff   128B 28 Oct 13:17 .
drwxr-x---+ 115 deepakadvani  staff   3.6K  6 Nov 00:07 ..
drwxr-xr-x   38 deepakadvani  staff   1.2K 28 Oct 13:17 fabric-samples
-rw-r--r--    1 deepakadvani  staff    11K 28 Oct 13:17 install-fabric.sh
(base) deepakadvani@deepaks-MacBook HyperledgerExperiment10 % ls -lah fabric-s
total 888
drwxr-xr-x  38 deepakadvani  staff   1.2K 28 Oct 13:17 .
drwxr-xr-x   4 deepakadvani  staff   128B 28 Oct 13:17 ..
-rw-r--r--   1 deepakadvani  staff   224B 28 Oct 13:17 .editorconfig
drwxr-xr-x  12 deepakadvani  staff   384B 28 Oct 13:17 .git
drwxr-xr-x   6 deepakadvani  staff   192B 28 Oct 13:17 .github
-rw-r--r--   1 deepakadvani  staff   510B 28 Oct 13:17 .gitignore
-rw-r--r--   1 deepakadvani  staff   707B 28 Oct 13:17 .golangci.yaml
drwxr-xr-x   4 deepakadvani  staff   128B 28 Oct 13:17 asset-transfer-abac
drwxr-xr-x  15 deepakadvani  staff   480B 28 Oct 13:17 asset-transfer-basic
drwxr-xr-x   9 deepakadvani  staff   288B 28 Oct 13:17 asset-transfer-events
drwxr-xr-x   7 deepakadvani  staff   224B 28 Oct 13:17 asset-transfer-ledger-q
drwxr-xr-x   8 deepakadvani  staff   256B 28 Oct 13:17 asset-transfer-private-
drwxr-xr-x   6 deepakadvani  staff   192B 28 Oct 13:17 asset-transfer-sbe
drwxr-xr-x   5 deepakadvani  staff   160B 28 Oct 13:17 asset-transfer-secured-
drwxr-xr-x   6 deepakadvani  staff   192B 28 Oct 13:17 auction-dutch
drwxr-xr-x   5 deepakadvani  staff   160B 28 Oct 13:17 auction-simple
drwxr-xr-x   8 deepakadvani  staff   256B 18 Jun 04:00 bin
-rw-r--r--   1 deepakadvani  staff   389K 28 Oct 13:17 CHANGELOG.md
drwxr-xr-x   3 deepakadvani  staff    96B 28 Oct 13:17 ci
-rw-r--r--   1 deepakadvani  staff   618B 28 Oct 13:17 CODE_OF_CONDUCT.md
-rw-r--r--   1 deepakadvani  staff   110B 28 Oct 13:17 CODEOWNERS
-rw-r--r--   1 deepakadvani  staff   935B 28 Oct 13:17 CONTRIBUTING.md
drwxr-xr-x  14 deepakadvani  staff   448B 28 Oct 13:17 full-stack-asset-transf
drwxr-xr-x   8 deepakadvani  staff   256B 28 Oct 13:17 hardware-security-modul
drwxr-xr-x   7 deepakadvani  staff   224B 28 Oct 13:17 high-throughput
-rw-r--r--   1 deepakadvani  staff    11K 28 Oct 13:17 LICENSE
-rw-r--r--   1 deepakadvani  staff   1.1K 28 Oct 13:17 MAINTAINERS.md
drwxr-xr-x   6 deepakadvani  staff   192B 28 Oct 13:17 off_chain_data
-rw-r--r--   1 deepakadvani  staff   8.3K 28 Oct 13:17 README.md
drwxr-xr-x   3 deepakadvani  staff    96B 28 Oct 13:17 test-application
drwxr-xr-x  17 deepakadvani  staff   544B 28 Oct 13:17 test-network
drwxr-xr-x   9 deepakadvani  staff   288B 28 Oct 13:17 test-network-k8s
drwxr-xr-x  34 deepakadvani  staff   1.1K 28 Oct 13:17 test-network-nano-bash
drwxr-xr-x   4 deepakadvani  staff   128B 28 Oct 13:17 token-erc-1155
drwxr-xr-x   6 deepakadvani  staff   192B 28 Oct 13:17 token-erc-20
drwxr-xr-x   6 deepakadvani  staff   192B 28 Oct 13:17 token-erc-721
drwxr-xr-x  21 deepakadvani  staff   672B 28 Oct 13:17 token-sdk
drwxr-xr-x   4 deepakadvani  staff   128B 28 Oct 13:17 token-utxo
(base) deepakadvani@deepaks-MacBook HyperledgerExperiment10 %
```

Step 3: Navigate to the test network

```
(base) deepakadvani@deepaks-MacBook HyperledgerExperiment10 % cd ~/HyperledgerExperiment10/fabric-samples/test-network
(base) deepakadvani@deepaks-MacBook test-network % ls
addOrg3                             compose                             network.config                      prometheus-grafana
fft-config                          configtx                            network.sh                          README.md
CHAINCODE_AS_A_SERVICE_TUTORIAL.md  monitordocker.sh                     organizations                       scripts
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 4A – 1.Check Fabric binaries

```
(base) deepakadvani@deepaks-MacBook test-network % ls -lah ../bin
total 313224
drwxr-xr-x   8 deepakadvani  staff   256B 18 Jun 04:00 .
drwxr-xr-x  38 deepakadvani  staff   1.2K 28 Oct 13:17 ..
-rwxr-xr-x   1 deepakadvani  staff    23M 18 Jun 04:00 configtxgen
-rwxr-xr-x   1 deepakadvani  staff    18M 18 Jun 04:00 cryptogen
-rwxr-xr-x   1 deepakadvani  staff    25M 18 Jun 04:00 discover
-rwxr-xr-x   1 deepakadvani  staff    26M 18 Jun 04:00 ledgerutil
-rwxr-xr-x   1 deepakadvani  staff    38M 18 Jun 04:00 orderer
-rwxr-xr-x   1 deepakadvani  staff    43M 18 Jun 04:00 peer
(base) deepakadvani@deepaks-MacBook test-network %
```

step 5 – Downloading and installing Hyperledger Fabric v2.5.16 binaries and Fabric CA client.

```
(base) deepakadvani@deepaks-MacBook HyperledgerExperiment10 % bash install-fabric.sh --fabric-version 2.5.16 binary

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.16 platform specific fabric binaries
===> Downloading:  https://github.com/hyperledger/fabric/releases/download/v2.5.16/hyperledger-fabric-darwin-arm64-2.5.16.tar.gz
===> Will unpack to: fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100  107M  100  107M    0     0  3328k      0  0:00:33  0:00:33 --:--:-- 3396k
==> Done.
===> Downloading version 1.5.17 platform specific fabric-ca-client binary
===> Downloading:  https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/hyperledger-fabric-ca-darwin-arm64-1.5.17.tar.gz
===> Will unpack to: fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 31.4M  100 31.4M    0     0  2874k      0  0:00:11  0:00:11 --:--:-- 2968k
==> Done.
(base) deepakadvani@deepaks-MacBook HyperledgerExperiment10 %
```

Step 6 – Start the Hyperledger Fabric network

```
(base) deepakadvani@deepaks-MacBook HyperledgerExperiment10 % cd ~/HyperledgerExperiment10/fabric-samples/test-network
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb' with crypto from 'cryptogen'
Unable to find image 'hyperledger/fabric-peer:latest' locally
latest: Pulling from hyperledger/fabric-peer
90a9d923273b: Pull complete
fff3795b4371: Pull complete
063722ae5baa: Pull complete
2a37fb9012ad: Pull complete
1c3fe66ec688: Pull complete
8d1585c202f2: Download complete
Digest: sha256:e07c735b6f2d131315f3ae6e582c5aa602143d9d14805bb98be8ebc9d6baa82c
Status: Downloaded newer image for hyperledger/fabric-peer:latest
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v3.1.5
Local fabric binaries and docker images are out of sync. This may cause problems.
/Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
[+] up 14/14
 ✔ Image hyperledger/fabric-orderer:latest Pulled                                                
```

Step 7 — Create the channel

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh createChannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb
Bringing up network
LOCAL_VERSION=v2.5.16
DOCKER_IMAGE_VERSION=v2.5.16
[+] up 3/3
 ✔ Container peer0.org2.example.com Running
 ✔ Container orderer.example.com    Running
 ✔ Container peer0.org1.example.com Running

 Generating channel genesis block 'mychannel.block'
 Using organization 1
 /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/../bin/configtxgen
 + '[' 0 -eq 1 ']'
 + configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychanne
 2026-11-06 00:41:36.035 IST 0001 INFO [common.tools.configtxgen] main -> Loading configuration
 2026-11-06 00:41:36.038 IST 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> order
 2026-11-06 00:41:36.038 IST 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Order
 max_inflight_blocks:5 snapshot_interval_size:16777216
 2026-11-06 00:41:36.038 IST 0004 INFO [common.tools.configtxgen.localconfig] Load -> Loaded configuration: /
 2026-11-06 00:41:36.043 IST 0005 INFO [common.tools.configtxgen] doOutputBlock -> Generating genesis block
 2026-11-06 00:41:36.043 IST 0006 INFO [common.tools.configtxgen] doOutputBlock -> Creating application chann
 2026-11-06 00:41:36.043 IST 0007 INFO [common.tools.configtxgen] doOutputBlock -> Writing genesis block
 + res=0
 Creating channel mychannel
 Adding orderers
 + . scripts/orderer.sh mychannel
 + '[' 0 -eq 1 ']'
 + res=0
 Status: 201
 {
         "name": "mychannel",
         "url": "/participation/v1/channels/mychannel",
         "consensusRelation": "consenter",
         "status": "active",
         "height": 1
 }

 Channel 'mychannel' created
 Joining org1 peer to the channel...
 Using organization 1
```



```
2026-11-06 00:41:48.404 IST 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org1MSP' on channel 'mychannel'
Setting anchor peer for org2...
Using organization 2
Fetching channel config for channel mychannel
Using organization 2
Fetching the most recent configuration block for the channel
++ peer channel fetch config /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/config_block.pb -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com
-c mychannel --tls --cafile /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem
2026-11-06 00:41:48.529 IST 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-11-06 00:41:48.536 IST 0002 INFO [cli.common] readBlock -> Received block: 1
2026-11-06 00:41:48.536 IST 0003 INFO [channelCmd] fetch -> Retrieving last config block: 1
2026-11-06 00:41:48.539 IST 0004 INFO [cli.common] readBlock -> Received block: 1
Decoding config block to JSON and isolating config to /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/Org2MSPconfig.json
++ configtxlator proto_decode --input /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/config_block.pb --type common.Block --output /Users/deepakadvani/Hyperledg
erExperiment10/fabric-samples/test-network/channel-artifacts/config_block.json
++ jq '.data.data[0].payload.data.config' /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/config_block.json
++ res=0
Generating anchor peer update transaction for Org2 on channel mychannel
++ jq '.channel_group.groups.Application.groups.Org2MSP.values += {"AnchorPeers":{"mod_policy": "Admins","value":{"anchor_peers": [{"host": "peer0.org2.example.com","port": 9051}]},"version": "0"}}' /Us
ers/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/Org2MSPconfig.json
++ res=0
++ configtxlator proto_encode --input /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/Org2MSPconfig.json --type common.Config --output /Users/deepakadvani/Hyper
ledgerExperiment10/fabric-samples/test-network/channel-artifacts/original_config.pb
++ configtxlator proto_encode --input /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/Org2MSPmodified_config.json --type common.Config --output /Users/deepakadv
ani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/modified_config.pb
++ configtxlator compute_update --channel_id mychannel --original /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/original_config.pb --updated /Users/deepakadva
ni/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/modified_config.pb --output /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/config_upda
te.pb
++ configtxlator proto_decode --input /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/config_update.pb --type common.ConfigUpdate --output /Users/deepakadvani/H
yperledgerExperiment10/fabric-samples/test-network/channel-artifacts/config_update.json
++ jq .
+++ cat /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/config_update.json
++ echo '{"payload":{"header":{"channel_header":{"channel_id":"mychannel", "type":2}},"data":{"config_update":{' '"channel_id":' '"mychannel",' '"isolated_data":' '{},' '"read_set":' '{' '"groups":' '{'
 '"Application":' '{' '"groups":' '{' '"Org2MSP":' '{' '"groups":' '{},' '"mod_policy":' '"",' '"policies":' '{' '"Admins":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Endorse
ment":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Readers":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Writers":' '{' '"mod_policy":' '"",' '"poli
cy":' null, '"version":' '"0"' '}' '},' '"values":' '{' '"MSP":' '{' '"mod_policy":' '"",' '"value":' null, '"version":' '"0"' '}' '},' '"version":' '"0"' '}' '},' '"mod_policy":' '"",' '"policies":' '{
},' '"values":' '{},' '"version":' '"0"' '}' '},' '"mod_policy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '},' '"write_set":' '{' '"groups":' '{' '"Application":' '{' '"groups":'
'{' '"Org2MSP":' '{' '"groups":' '{},' '"mod_policy":' '"Admins",' '"policies":' '{' '"Admins":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Endorsement":' '{' '"mod_policy":'
'"",' '"policy":' null, '"version":' '"0"' '},' '"Readers":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0"' '},' '"Writers":' '{' '"mod_policy":' '"",' '"policy":' null, '"version":' '"0
"' '}' '},' '"values":' '{' '"AnchorPeers":' '{' '"mod_policy":' '"Admins"," '"value":' '{' '"anchor_peers":' '[' '{' '"host":' '"peer0.org2.example.com"," '"port":' 9051 '}' ']' '},' '"version":' '"0"'
 '},' '"MSP":' '{' '"mod_policy":' '"",' '"value":' null, '"version":' '"0"' '}' '},' '"version":' '"1"' '}' '},' '"mod_policy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '}' '},'
'"mod_policy":' '"",' '"policies":' '{},' '"values":' '{},' '"version":' '"0"' '}' '}}}}'
++ configtxlator proto_encode --input /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/config_update_in_envelope.json --type common.Envelope --output /Users/deep
akadvani/HyperledgerExperiment10/fabric-samples/test-network/channel-artifacts/Org2MSPanchors.tx
2026-11-06 00:41:48.708 IST 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-11-06 00:41:48.729 IST 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org2MSP' on channel 'mychannel'
Channel 'mychannel' joined
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 8 — Check the Go chaincode folder

```
(base) deepakadvani@deepaks-MacBook test-network % ls ../asset-transfer-basic/chaincode-go
assetTransfer.go        chaincode               go.mod                  go.sum
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 9 — Deploy the smart contract

```
hecking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'
sing organization 2
 peer lifecycle chaincode approveformyorg -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/orga
zations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --version 1.0 --package-id basic_1.0:d0b27e1af09f3039dabac82678c64e8807311f9405008efb867ba8
5dd643aa --sequence 1
 res=0
026-11-06 00:53:22.761 IST 0001 INFO [chaincodeCmd] ClientWait -> txid [d5d0042446342ff63a3bd380a055210d2fc799d63119d1c616cd36e4ea6b2766] committed with status (VALID) at localhost:9051
haincode definition approved on peer0.org2 on channel 'mychannel'
sing organization 1
hecking the commit readiness of the chaincode definition on peer0.org1 on channel 'mychannel'...
ttempting to check the commit readiness of the chaincode definition on peer0.org1, Retry after 3 seconds.
 peer lifecycle chaincode checkcommitreadiness --channelID mychannel --name basic --version 1.0 --sequence 1 --output json
 res=0

      "approvals": {
              "Org1MSP": true,
              "Org2MSP": true
      }

hecking the commit readiness of the chaincode definition successful on peer0.org1 on channel 'mychannel'
sing organization 2
hecking the commit readiness of the chaincode definition on peer0.org2 on channel 'mychannel'...
ttempting to check the commit readiness of the chaincode definition on peer0.org2, Retry after 3 seconds.
 peer lifecycle chaincode checkcommitreadiness --channelID mychannel --name basic --version 1.0 --sequence 1 --output json
 res=0

      "approvals": {
              "Org1MSP": true,
              "Org2MSP": true
      }

hecking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'
sing organization 1
sing organization 2
 peer lifecycle chaincode commit -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/
rdererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --peerAddresses localhost:7051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/fabric-
amples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/
abric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem --version 1.0 --sequence 1
 res=0
026-11-06 00:53:31.337 IST 0001 INFO [chaincodeCmd] ClientWait -> txid [5489dd678d4e46647897a53ad821e91f95c916b7fec93ec2fdde389243ef7487] committed with status (VALID) at localhost:9051
026-11-06 00:53:31.348 IST 0002 INFO [chaincodeCmd] ClientWait -> txid [5489dd678d4e46647897a53ad821e91f95c916b7fec93ec2fdde389243ef7487] committed with status (VALID) at localhost:7051
haincode definition committed on channel 'mychannel'
sing organization 1
uerying chaincode definition on peer0.org1 on channel 'mychannel'...
[Attempting to Query committed status on peer0.org1, Retry after 3 seconds.
 peer lifecycle chaincode querycommitted --channelID mychannel --name basic
 res=0
ommitted chaincode definition for chaincode 'basic' on channel 'mychannel':
ersion: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
uery chaincode definition successful on peer0.org1 on channel 'mychannel'
sing organization 2
uerying chaincode definition on peer0.org2 on channel 'mychannel'...
ttempting to Query committed status on peer0.org2, Retry after 3 seconds.
 peer lifecycle chaincode querycommitted --channelID mychannel --name basic
 res=0
ommitted chaincode definition for chaincode 'basic' on channel 'mychannel':
ersion: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
uery chaincode definition successful on peer0.org2 on channel 'mychannel'
haincode initialization is not required
base) deepakadvani@deepaks-MacBook test-network %
```

Step 10 — Initialize the Ledger

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh cc invoke -c mychannel -ccn basic -ccic '{"Args":["InitLedger"]}'
Using docker and docker-compose
Using organization 1
Invoking on peer0.org1 on channel 'mychannel'...
usage: sleep number[unit] [...]
Unit can be 's' (seconds, the default), m (minutes), h (hours), or d (days).
Attempting to Invoke on peer0.org1, Retry after  seconds.
+ peer chaincode invoke -o localhost:7050 -C mychannel -n basic -c '{"Args":["InitLedger"]}' --tls --cafile /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizati
rganizations/example.com/tlsca/tlsca.example.com-cert.pem --peerAddresses localhost:7051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizati
nizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/org
eerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem
+ res=0
2026-11-06 01:00:29.679 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
Invoke successful on peer0.org1 on channel 'mychannel'
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 11 — Read all assets from the ledger

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh cc query -c mychannel -ccn basic -ccqc '{"Args":["GetAllAssets"]}'
Using docker and docker-compose
Using organization 1
Querying on peer0.org1 on channel 'mychannel'...
usage: sleep number[unit] [...]
Unit can be 's' (seconds, the default), m (minutes), h (hours), or d (days).
Attempting to Query peer0.org1, Retry after  seconds.
+ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
+ res=0
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5},{"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10},{"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10},{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15},{"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15}]
Query successful on peer0.org1 on channel 'mychannel'
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 12 — Create a new asset

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh cc invoke -c mychannel -ccn basic -ccic '{"Args":["CreateAsset","asset7","purple","20","Deepak","900"]}'
Using docker and docker-compose
Using organization 1
Invoking on peer0.org1 on channel 'mychannel'...
usage: sleep number[unit] [...]
Unit can be 's' (seconds, the default), m (minutes), h (hours), or d (days).
Attempting to Invoke on peer0.org1, Retry after  seconds.
+ peer chaincode invoke -o localhost:7050 -C mychannel -n basic -c '{"Args":["CreateAsset","asset7","purple","20","Deepak","900"]}' --tls --cafile /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --peerAddresses localhost:7051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem
+ res=0
2026-11-06 01:04:05.057 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
Invoke successful on peer0.org1 on channel 'mychannel'
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 13 — Read the newly created asset

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh cc query -c mychannel -ccn basic -ccqc '{"Args":["ReadAsset","asset7"]}'
Using docker and docker-compose
Using organization 1
Querying on peer0.org1 on channel 'mychannel'...
usage: sleep number[unit] [...]
Unit can be 's' (seconds, the default), m (minutes), h (hours), or d (days).
Attempting to Query peer0.org1, Retry after  seconds.
+ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset7"]}'
+ res=0
{"AppraisedValue":900,"Color":"purple","ID":"asset7","Owner":"Deepak","Size":20}
Query successful on peer0.org1 on channel 'mychannel'
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 14 — Update an existing asset

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh cc invoke -c mychannel -ccn basic -ccic '{"Args":["UpdateAsset","asset7","blue","25","Deepak","1200"]}'
Using docker and docker-compose
Using organization 1
Invoking on peer0.org1 on channel 'mychannel'...
usage: sleep number[unit] [...]
Unit can be 's' (seconds, the default), m (minutes), h (hours), or d (days).
Attempting to Invoke on peer0.org1, Retry after  seconds.
+ peer chaincode invoke -o localhost:7050 -C mychannel -n basic -c '{"Args":["UpdateAsset","asset7","blue","25","Deepak","1200"]}' --tls --cafile /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --peerAddresses localhost:7051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem
+ res=0
2026-11-06 01:06:57.147 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200
Invoke successful on peer0.org1 on channel 'mychannel'
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 15 — Verify the updated asset

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh cc query -c mychannel -ccn basic -ccqc '{"Args":["ReadAsset","asset7"]}'
Using docker and docker-compose
Using organization 1
Querying on peer0.org1 on channel 'mychannel'...
usage: sleep number[unit] [...]
Unit can be 's' (seconds, the default), m (minutes), h (hours), or d (days).
Attempting to Query peer0.org1, Retry after  seconds.
+ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset7"]}'
+ res=0
{"AppraisedValue":1200,"Color":"blue","ID":"asset7","Owner":"Deepak","Size":25}
Query successful on peer0.org1 on channel 'mychannel'
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 16 — Transfer asset ownership

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh cc invoke -c mychannel -ccn basic -ccic '{"Args":["TransferAsset","asset7","Rahul"]}'
Using docker and docker-compose
Using organization 1
Invoking on peer0.org1 on channel 'mychannel'...
usage: sleep number[unit] [...]
Unit can be 's' (seconds, the default), m (minutes), h (hours), or d (days).
Attempting to Invoke on peer0.org1, Retry after  seconds.
+ peer chaincode invoke -o localhost:7050 -C mychannel -n basic -c '{"Args":["TransferAsset","asset7","Rahul"]}' --tls --cafile /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --peerAddresses localhost:7051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /Users/deepakadvani/HyperledgerExperiment10/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem
+ res=0
2026-11-06 01:09:12.240 IST 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payload:"Deepak"
Invoke successful on peer0.org1 on channel 'mychannel'
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 17 — Verify ownership

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh cc query -c mychannel -ccn basic -ccqc '{"Args":["ReadAsset","asset7"]}'
Using docker and docker-compose
Using organization 1
Querying on peer0.org1 on channel 'mychannel'...
usage: sleep number[unit] [...]
Unit can be 's' (seconds, the default), m (minutes), h (hours), or d (days).
Attempting to Query peer0.org1, Retry after  seconds.
+ peer chaincode query -C mychannel -n basic -c '{"Args":["ReadAsset","asset7"]}'
+ res=0
{"AppraisedValue":1200,"Color":"blue","ID":"asset7","Owner":"Rahul","Size":25}
Query successful on peer0.org1 on channel 'mychannel'
(base) deepakadvani@deepaks-MacBook test-network %
```

Step 18 — Display all assets

```
(base) deepakadvani@deepaks-MacBook test-network % ./network.sh cc query -c mychannel -ccn basic -ccqc '{"Args":["GetAllAssets"]}'
Using docker and docker-compose
Using organization 1
Querying on peer0.org1 on channel 'mychannel'...
usage: sleep number[unit] [...]
Unit can be 's' (seconds, the default), m (minutes), h (hours), or d (days).
Attempting to Query peer0.org1, Retry after  seconds.
+ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
+ res=0
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5},{"AppraisedValue":400,"Color":"red","ID":"asset2","Owner":"Brad","Size":5},{"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10},{"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10},{"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15},{"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15},{"AppraisedValue":1200,"Color":"blue","ID":"asset7","Owner":"Rahul","Size":25}]
Query successful on peer0.org1 on channel 'mychannel'
(base) deepakadvani@deepaks-MacBook test-network %
```

Conclusion : Hyperledger Fabric network was successfully established, and the asset-transfer-basic smart contract was deployed on the `mychannel` channel. Asset creation, reading, updating and ownership transfer operations were performed successfully. The final ledger query verified all seven assets.